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Hierarchical Coloured Petri-Net Based Multi-Agent System for Flood Monitoring, Prediction, and Rescue (FMPR)

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ABSTRACT In Punjab Pakistan, a river flood event is the most important natural disasters that every year causes high human casualties as well as heavy economic losses. A safety-critical system should be correct, reliable, complete, consistent, and unambiguous for accurate and precise flood monitoring, prediction, as well as emergency and rescue services during or after a flood event. System correctness must be ensured in every phase of the development of a safety-critical Flood Monitoring, Prediction, and Rescue (FMPR) system. Correctness is a functional property related to the behavior of the system having sub-properties of safety and liveness. These sub-properties must also be correct, complete, consistent, and unambiguous. Rigorous mathematical based software engineering methods like formal methods are suitable for the specification, design, modeling, verification, and validation of an FMPR safety-critical system. The proposed safety-critical system for FMPR is distributed as it is based on multi-agents. This system is specified, analyzed and designed by using Gaia multi-agent methodology, which is based on organizational abstractions. Gaia role model-based agent roles are specified, regular expression based liveness properties are specified, and first-order predicate calculus based safety properties are specified. Thus, safety and liveness properties are formally specified. After the detailed design phase, the system is formally modeled and verified by Hierarchical Coloured-Petri Nets (CP-Nets). As a result, a novel formal system for the specification, analysis, design, modeling, and verification of the FMPR system is proposed. This safety-critical system is distributed based on multi-agents.

INDEX TERMS Flood monitoring, prediction, and rescue (FMPR), formal specification, system modelling, correctness, safety property, liveness property, hierarchical coloured-Petri nets (CP-Nets).

I. INTRODUCTION

Flood is a natural disaster event that affects millions of people each year in Punjab province, Pakistan. It results in huge volumes of water, much more than the natural or artificial conveyance system (i.e. channels, canals, streams, rivers, dams, creeks, basins, culverts, and estuaries) can manage. In Punjab, climate change is the major cause of floods resulting in the rapid melting of snow in northern

Himalayan mountains, and excessive monsoon rains. The flood damages can be reduced by modern-day technological and scientific advancements. Modern-day early warning systems can inform stakeholders in time, rescue system can inform stakeholders about rescue services (i.e. shelter, clean drinking water, cooked food, medical facilities, medicines, etc). In Punjab province, poverty results in poorly planned illegal human settlements in flood risk areas (i.e. floodplains adjacent to riverbanks, areas not protected by flood banks). These settlements result in high casualties and losses during floods.

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According to the official journal of European union [1], the directive 2007/60/EC of the European parliament and of the council of 23 October 2007 on the assessment and management of flood risks, flood cause displacement of people, human fatalities, damage to crops, fatalities of livestock animals, and halt the economic activities and economic development. The increase in human population, increase in human settlements in flood risk plains, the development of economic assets in flood plains, climate change, the reduction of natural water retention of land, increase the risks and adverse impact of floods. Flood risk management plays a critical role in the proposition of an approach for the management of flood risks at the community level.

To model, design, and develop a formal distributed system based on multi-agents, to overcome the flood hazard, to provide rescue services, it is fundamental to study the principal scientific phenomenon involved in the generation of floods. Reference [2] have presented studies about the scientific phenomenon involved in the generation of floods. These studies help in predicting the future behavior of floods. Reference [3] have categorized the water-related issues into three groups: the excessive presence of water, water availability, and water scarcity. Reference [4] have studied the flash flood risks and inundation phenomenon in the western part of Saudi Arabia. In this region, there are thousands of people living in low-lying regions with a high risk of flash floods and inundation. This study has helped us in our work of analyzing, designing, modeling, and implementing an FMPR system in Punjab. Reference [5] examines the relationship of philosophy, logic, and science to engineering i.e. the relevance of philosophy of science and the importance of fuzzy logic to engineering. The modeling principles and philosophy in engineering are examined and discussed. This work promotes the recognition of artistic and humanitarian aspects and their integration in engineering works, in addition to readily available formulation and software. Therefore this work promotes philosophical, humanitarian, logic aspects for the design and development of flood monitoring, prediction, and rescue systems.

According to [6], over the last 37 years, Pakistan has faced 17 floods of different intensities after the Mega-flood in 1929. The floods in 2010 affected millions of people in Punjab, Pakistan [6]. Flood inundation maps, early warning flood plans are important for flood-risk regions protection. The community must be provided with education about flood awareness. The flood inundation maps have information about several very important factors like the low-lying high-risk areas, the damaged flood banks, the important high-priority areas that should be protected on all costs, the flood banks that should be blown in case of high floods, to inundate the low-risk low-priority non-populated regions to protect the high-priority high-populated regions. These maps are important and should be developed and made available to the community.

Punjab is an agriculture-dependent province; there are five major rivers and tens of canals originating from these rivers. The role that water plays for agriculture is fundamental

and without an effective river and canal system, there is no agriculture. Historical facts and figures have shown that Pakistan is in the list of countries that face severe floods and millions of people's lives are affected as a result of no prior flood-intimation, lack of prior-information and poor rescue services.

There is a need for new dams, water reservoirs, canals, lakes to store fresh water for agriculture and drinking use. Currently, large quantities of fresh water are wasted (i.e. that flows through rivers and falls into the Indian ocean. These water reserves are important to cope with climate change. An effective prevention strategy can minimize the destruction from floods. Large quantities of flood data are collected from flood perceivers, and advanced prediction and forecasts, based on this flood data, are useful to calculate with accuracy the probability of the flood.

In semi-arid and arid regions, the floods are also beneficial events, their benefits are:

- 1) Flood water is the only source of groundwater recharge. In cities and towns near the river the groundwater recharges, its quality improves, and its level rises. The aquifers are re-filled and their water quality improves.
- 2) Flood water is also the most important source of enrichment of agriculture lands with important minerals. This fertile soil yields better crops for the next few years, which is beneficial for the agriculture economy of Punjab.
- 3) In some regions of Punjab, there are run-off floodwater harvesting (i.e. harvesting crops in agriculture lands flooded by floodwater).
- 4) The floodwater also re-fills the fresh water ponds, lakes, dams, and water reservoirs.
- 5) South Punjab is among the hottest regions of the world. In these regions after a flood, water evaporation increases, which in turn causes rain. The overall weather becomes cooler and these rains are extremely beneficial for these hot, low-rain arid regions.

In semi-arid and arid regions with mountains and valleys, flash floods are the most dangerous disaster. Flash floods occur in those semi-arid and arid regions where there is high-intensity rainfall in short periods, very few vegetation and steep topography. In most cases, these high-risk regions are the low-lying areas surrounded by high mountains and adjacent to flood plains. Sudden inundation risks the downstream sections. Low-lying areas suffer most from floods and inundation hazards. In Punjab, thousands of people live in those low-lying areas because of cheap housing, cheap land, and the availability of groundwater. Floods can also arise due to dam failure or dam destruction due to earthquakes or terrorism.

- 1) **Heavy monsoon rains** generate intensive storm and floods in Punjab, which increase the water-level in rivers and canals.
- 2) **Rapid snow melting** in the northern glaciers and mountains of Himalaya and Hindu-Kush due to climate change

has increased the water in the rivers and canals, thus increasing the risk of floods each year.

The occurrences and magnitude of floods have increased in Punjab over the last fifteen years. Large parts of Punjab are arid and semi-arid, and due to climate change, there are extensive rains that have resulted in several flooding events. The losses continue to increase, due to the following two reasons:

- 1) **Human reasons**, due to an increase in the vulnerability of flood-prone areas as a result of extensive human habitation. The mismanagement, poor organization, and corruption in government decision-making departments have resulted in the construction of infrastructure and buildings in flood-prone areas.
- 2) **Physical reasons**, related to global warming, rapid melting of snow in northern Himalayan glaciers, increase in the magnitude and frequency of rainfall and consequent flood events [7].

The above two factors are important for the understanding of flood risks, flood inundation maps, flood monitoring, prediction, and rescue operations. The human-related factors are the major cause of floods. During the monsoon season in Punjab, the rainfall varies in each region and it is erratic temporally and spatially. In some regions, an individual rainfall during a storm may exceed the mean annual rainfall leading to floods. Reference [3] have studied high-intensity rainfalls, erratic rainfalls, and rainfalls during a storm in the arid regions of Saudi Arabia.

It is important to calculate factors that cause floods and specify the safety and liveness properties of the FMPR system. The safety properties specify all those aspects of the system that highlight the boundary conditions, constraints, and limitations of the system. It is important to calculate rainfall for flood risk assessment. To calculate annual mean rainfall the impact of climate change must be taken into consideration. Without studying the impact of climate change, effective engineering and modeling of water resources cannot be done. Flood safety analysis helps to build safe infrastructures for retaining the floodwater volume, passing flood water through different mediums (i.e. natural channels, man-made channels, flood dams, man-made dams, rivers, canals, etc) in a controlled manner with the maintenance of safety and stability in every aspect. Safety property is of particular importance in a safety-critical computer system [8] like an FMPR system. In safety standards such as IEC61508 [9] and EN50128 [10], formal methods are strongly recommended for software requirements specification and software design. Figure 1 depicts all the stakeholders of the FMPR system.

As shown in figure 1, the meteorological department, disaster management department, flood perceivers, social services department, and independent observers provide important data to the FMPR system, which processes this data into useful information. Based on this information flood warning alerts, real-time flood alerts, flood prevention strategies,

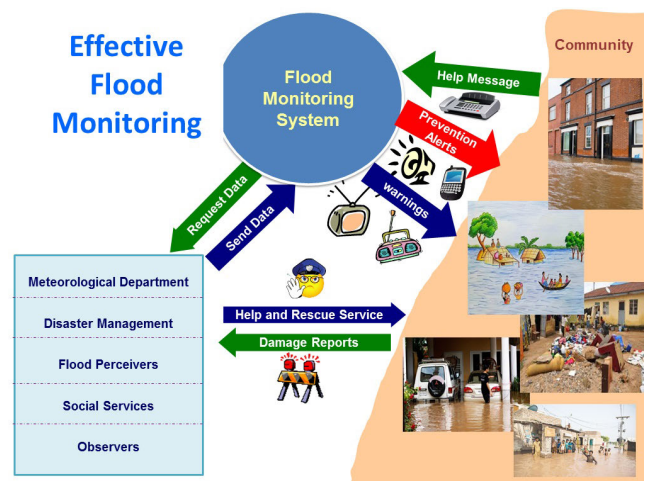


FIGURE 1. Stakeholders of the FMPR system.

flood damage reports, flood inundation maps, flood help services, and flood rescue services are performed.

A. CONTRIBUTIONS

FMPR system is a safety-critical system, the application of formal methods and techniques for the analysis, design, modeling, verification, and validation of a safety-critical system is an important area of research in academia and industry. In this work, our contributions are:

- 1) In this system, failure can be potentially disastrous, and correctness is of fundamental importance. A correct and reliable system is important, and if the system is not able to warn people on time then it is of no use.
- 2) We have analyzed, designed, modeled, and verified a multi-agent model of an FMPR system for the monitoring of rivers, canals and dams before, during, and after floods; and rescue and emergency services in a post-flood disaster situation. A model of the system is created, model-checked, and verified by mathematical proofs.
- 3) The requirement specification, analysis, and design of a FMPR system based on multi-agents. It is a distributed system and it works even if several agents fail, therefore it is fault-tolerant.
- 4) *Formal specification and design* constructed by using Gaia multi-agent methodology. The Gaia's role model plays the fundamental role in formal specification of liveness (i.e. using regular expression) and safety property (i.e. using first-order predicate calculus).
- 5) It provides *formal modeling* by using Hierarchical CP-Nets.
- 6) It provides *formal verification of correctness properties* by using Hierarchical CP-Nets based model checking.
- 7) *Hierarchical CP-Nets* is used, with two levels of abstractions. As a result of this two-level refinement, the model of the flood monitoring system is modular (i.e. broken down into well-defined sub-parts). This modular system has a high level of human-machine interaction and is more user-friendly. This two-layer model helps in exhaustive model-checking.

- 8) The system provides *emergency and rescue services* after a flood. If the major communication infrastructure is destroyed by the disaster, the alternative communication mediums (i.e. radio, internet, social media, electronic media, mobile network, etc.) if the major communication infrastructure is destroyed by the disaster.
- 9) The system is centered on *uninterrupted continuous communication* between the FMPR system and stakeholders (i.e. law and order authorities, rescue services, affected population, social media, electronic media, and government authorities). Propagation of information to these stakeholders ensures early action after flood alerts. CP-Nets model of communications is constructed and model-checked.

II. MOTIVATION AND PROBLEM STATEMENT

There is no existing approach for the design, modeling, and verification of an FMPR system that is based on multi-agents and supports formal analysis, modeling, and verification of liveness and safety properties.

A proposed multi-agent-based FMPR system is a safety-critical system. Correctness is important as errors and safety property violations of this system can cause human life loss and high economic losses. This FMPR system has to operate in an open environment and it has to continuously interact and collaborate with other systems, thus using the traditional automated testing method for verification is not sufficient. Therefore, for the specification, modeling, and verification of a correct safety-critical system, we have used formal methods and techniques.

Problem Statement: Each year floods results in the loss of human lives, loss of human habitat, loss of agriculture, and displacement of hundreds of thousands of people. The problematic is the analysis, design, and modeling of a correct, reliable safety-critical system for FMPR based on multi-agents, to model liveness and safety properties of an FMPR system. It must be ensured that these liveness and safety properties of an FMPR system are correct and reliable.

III. RESEARCH OBJECTIVES AND RESEARCH QUESTION

- 1) Analysis, design, modeling of an FMPR system based on multi-agents.
- 2) Formal verification of the correctness properties of liveness and safety of the FMPR system.
- 3) Modeling of rescue services in flood-affected areas during and after floods.
- 4) Analysis, preliminary and detailed design of the multi-agent-based FMPR system by using organizational abstractions and role models of the Gaia multi-agent methodology.
- 5) Formal verification of the multi-agent-based FMPR system by using hierarchical CP-Nets.

Research Question: *How is it possible to propose a novel approach based on multiple state-of-the-art formal methods and techniques for the analysis, preliminary design,*

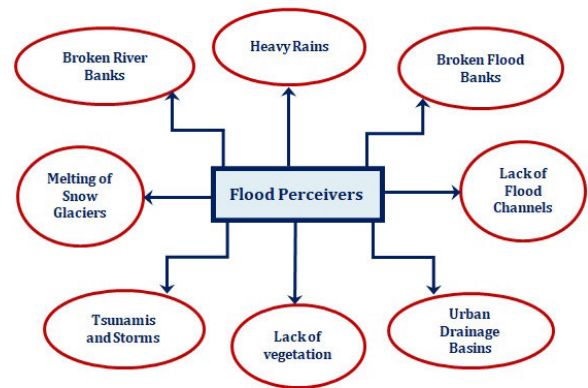


FIGURE 2. Flood monitoring perceivers.

detailed design, formal verification, formal modeling, and model checking of a multi-agent FMPR system.

IV. LITERATURE REVIEW

A. FLOOD MONITORING PERCEIVERS, SENSORS, AND ACTUATORS

Flood Monitoring perceivers shown in Figure 2, are used for early detection of floods as well as the analysis of flood intensity.

These perceivers have multiple sensors that identify the factors that cause floods. These sensors identify damaged flood banks, heavy rains, over-flowing rivers, rate of snow-melting in northern Himalayan glaciers, lack of trees and vegetation near river and canal banks, alerts for tsunamis, alerts for storms, alerts for earthquakes, urban drainage basins.

Flood perceivers data are of critical importance for the FMPR system. Based on this data the future predictions and decisions are made, which are vital for saving humans' lives and property. There is a perceiver that calculates the rate of snow melting in northern glaciers. This perceiver estimates climate change. As the rate of snow melting in northern glaciers is directly proportional to the rise in temperature. During the last few years, the rate of snow melting has considerably increased. These flood monitoring perceivers are an important component of the FMPR system. After a feasibility study, the requirements are identified, then Gaia multi-agent [11], [12] detailed role model is constructed for a better understanding of the flood monitoring perceivers, and then in the subsequent stage, formal models are constructed. A CP-Nets centered formal model is created and model-checked, then the correctness properties of safety and liveness are ensured through CP-Nets model checking.

The requirements are elicited according to the functionality of the flood monitoring perceivers. The functional and non-functional requirements are specified for all users of the system. Flood monitoring, prediction, and rescue based on flood perceivers is an iterative procedure that involves all the stakeholders i.e. government flood monitoring agencies, disaster management departments, rescue services, hospitals, schools, mosques, and flood-area population.

The flood perceivers directly influence flood prevention as well as after-flood rescue services. The primary goal is to build a correct formally verified model.

B. AGENT-BASED MODELING

Agent-based modeling is a distributed modeling technique well-suited to specify, model, and verify systems ranging from research to industrial issues [13]–[16]. Agent-Based modeling helped the researchers and developers to propose several analysis, design and development techniques. Some of these techniques are centered on models (i.e. use, creation, and validation). Comprehensive models are analyzed extensively in [12] and [17]–[21].

1) GAIA MULTI-AGENT METHODOLOGY

Gaia multi-agent methodology [11], [12] is ideal for the analysis, preliminary design, detailed design of a multi-agent system based on organizational abstractions and models (i.e. organizational model, role model, interaction model). The organizational rules also play a central role in Gaia. We have formalized the Gaia role model based liveness and safety properties of correctness. In our proposed approach liveness property is specified in regular expression and safety is specified in first-order predicate calculus. Safety and liveness are formal constructs that are specified by using regular expression and first-order predicate logic respectively. These two formal properties should be complete. These two formal constructs are easy to use and are expressive to specify all types of properties exhaustively.

C. FORMAL MODELING AND VERIFICATION

Formal modeling and verification are carried out by using methods and techniques having a sound mathematical foundation. Formal methods are used in requirement specification and design level to ensure correctness properties by using state transition machines, abstraction principle, refinement principle, proof obligations, theorem proving and inductive assertions [19]. These methods and techniques provide a precise, accurate, and rigorous foundation for the requirement specification, design, modeling, and verification. In complicated systems like a safety-critical FMPR system, multiple formal techniques can be applied at each development stage, to ensure rigorous verification. Formal verification is exhaustive, rigorous, and it ensures the correctness properties of liveness and safety of the system.

As shown in Table 1, the specifications within the analysis and design phases of the Gaia's role model are formal, the safety properties are specified by using First-Order Predicate Calculus and liveness properties are specified by using Regular Expressions. Both of these constructs are mathematical based and thus formal. Formal modeling and verification by model checking are implemented by the Hierarchical CP-Nets model. A formal CP-Nets model of the system is developed with precisely defined functional as well as non-functional properties. A model provides a basis to improve and verify the system in a rigorous manner, it is also important

in safeguarding correctness of the system. This model takes user input and systemically checks whether or not the correctness properties of liveness and safety are ensured by the system.

D. MODEL CHECKING

Model checking [22]–[29] is a practical formal method for automatic, systematic, and exhaustive verification. Its objective is verification by specifying and automated checking of correctness properties. A mathematical model of the system is created and a comprehensive review of the model is performed. It includes checking all states and transitions in the model i.e. exhaustive analysis of the model. It creates an abstract representation of a system with all the possible states and transitions. The correctness of the model is verified through model checking techniques. The model takes input and systemically checks whether the system properties are ensured by the system. It is widely used in verification of safety-critical systems i.e. verification of software systems of airplanes, spacecraft, trains, metros, satellites, control units of a nuclear reactor, and monitoring systems for earthquakes, tsunamis, floods, etc.

Model checking uses temporal logic [30], linear-time temporal logic (LTL) [31]–[33] and computation-tree temporal logic (CTL) [30], [34], [35], for stating, checking, and verifying behavioral properties.

According to [26] model checking is an automated technique, a formal property is defined, a finite-state model of a system is constructed, then the automated tool model checker automatically and systematically evaluates and verifies that this formal property holds for the model. Model-checking is based on models and in these model-based formal verification techniques the system behavior is described in a mathematical-based unambiguous and precise specifications. As compared to formal specifications, informal and semi-formal system specifications are inconsistent, ambiguous, and incomplete. The model-checker implements an algorithm that systematically explores all possible states of the model (i.e. exhaustive exploration of state space). Therefore, model-checker gives exhaustive verification and validation as compared to simulation (i.e. restrictive set of scenarios of model) and testing (i.e. restrictive set of test-cases of system implementation)

E. CORRECTNESS PROPERTIES

Correctness properties of safety and liveness provide exhaustive system verification. Safety and liveness properties both are equally important for the formal verification, they complement each other (i.e. liveness alone or safety alone is not sufficient to ensure correctness and reliability). These are both functional properties.

The safety property is an invariant which asserts that “something bad never happens, that an acceptable state of affairs is maintained”. According to [36] safety property “ $S = a_1, a_2 \dots a_n$ is a deterministic process that asserts that any trace including actions in the alphabet of S, is accepted

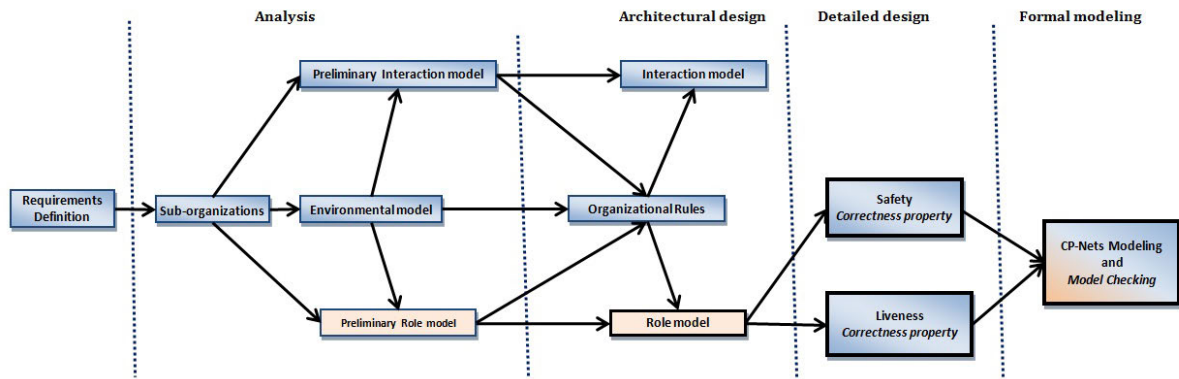


FIGURE 3. Safety and liveness properties of Gaia multi-agent role model and their CP-nets modeling.

TABLE 1. Gaia multi-agent methodology for analysis, preliminary design, and detailed design and CP-nets for formal modeling and model checking.

Analysis	Preliminary design	Detailed design	Formal modeling
<i>Gaia</i>	<i>Gaia</i>	<i>Gaia</i>	<i>CP-Nets</i>
Organizational model Environmental model	Roles model Interaction model	Safety Property: (First-Order Predicate Calculus) Liveness Property: (Regular Expression)	CP-Nets Model Checking
Preliminary roles model Preliminary interaction model Organizational rules			Safety and Liveness properties modeled in CP-Nets

by S. ERROR conditions are like exceptions which state what is not required, as in complex systems we specify safety properties by directly stating what is required”. According to [37] “a safety property is a property that can be specified by a safety formula of the form $\Box p$ (i.e. temporal operator $\Box$ meaning always). This formula states that the property p holds throughout the computation”.

The liveness property asserts that “something good happens which describes the states of a system that bring about given certain conditions” [18]. The liveness property is directly related to the core behavior of the system. The functions and roles of the system are specified as liveness. On the other hand, safety property is used to specify the constraints under which the system has to operate. The quality attributes of the system can also be specified as safety properties. The safety and liveness properties of the system are ensured by formally specifying and verifying these properties in CP-Nets.

F. COLOURED-PETRI NETS (CP-NETS)

A CP-Nets [38]–[43] is a formal, discrete mathematics-based graphical language that constructs an executable model of the system. This model consists of places, transitions, and tokens flowing between places through transitions. “A non-hierarchical Coloured Petri Net is a nine-tuple $CP-Net = (P, T, A, \Sigma, V, C, G, E, I)$, where:

- 1) P is a finite set of places.
- 2) T is a finite set of transitions T such that $P \cap T = \phi$.

- 3) $A \subseteq P \times T \cup T \times P$ is a set of directed arcs
- 4) Σ is a finite set of non-empty colour sets.
- 5) V is a finite set of typed variables such that $Type[v] \in \Sigma$ for all variables $v \in V$.
- 6) $C: P \rightarrow \Sigma$ is a colour set function that assigns a colour set to each place.
- 7) $G: T \rightarrow EXPR V$ is a guard function that assigns a guard to each transition t such that $Type[G(t)] = Bool$.
- 8) $E: A \rightarrow EXPR V$ is an arc expression function that assigns an arc expression to each arc a such that $Type[E(a)] = C(p)MS$, where p is the place connected to the arc a .
- 9) $I: P \rightarrow EXPR \phi$ is an initialization function that assigns an initialization expression to each place p such that $Type[I(p)] = C(p)MS$ ” [39].

CP-Nets can be compiled and executed. They are ideal to model the behavior of real-time safety-critical systems. A safety-critical system can be modeled in the form of CP-Nets and its correctness properties of safety and liveness can be exhaustively studied and verified. The specifications in CP-Net are represented in both textual form (i.e. ML of CP-Nets), and diagrammatic form (i.e. places, transitions, and arcs). It exhaustively verifies a system, in which concurrency, communications, and synchronizations play the key role.

G. HIERARCHICAL COLOURED-PETRI NETS (CP-NETS)

Hierarchical CP-Net has multiple layers (i.e. abstraction levels) of CP-Nets. It has a modular design as compared to simple CP-Net. Large complicated and complex systems

are modeled by using hierarchical CP-Nets. “Hierarchical CP-Net is a four-tuple $CPN_H = (S, SM, PS, FS)$ where:

- 1) S is a finite set of modules. Each module is a Coloured Petri Net Module $s = ((P^s, T^s, A^s, \Sigma^s, V^s, C^s, G^s, E^s, I^s), T_{sub}^s, P_{port}^s, PT^s)$. It is required that $(P^{s1} \cup T^{s1}) \cap (P^{s2} \cup T^{s2}) = \emptyset$ for all $s1, s2 \in S$ such that $s1 \neq s2$.
- 2) $SM: T_{sub} \rightarrow S$ is a sub-module function that assigns a sub-module to each substitution transition. It is required that the module hierarchy is acyclic.
- 3) PS is a port-socket relation function that assigns a port-socket relation $PS(t) \subseteq P_{sock}(t) \times P_{port}^{M(t)}$ to each substitution transition t . It is required that $ST(p) = PT(p^c)$, $C(p) = C(p^c)$, and $I(p) \langle \rangle = I(p^c) \langle \rangle$ for all $(p, p^c) \in PS(t)$ and all $t \in T_{sub}$.
- 4) $FS \subseteq 2^P$ is a set of non-empty fusion sets such that $C(p) = C(p^c)$ and $I(p) \langle \rangle = I(p^c) \langle \rangle$ for all $p, p^c \in fs$ and all $fs \in FS$ ” [39].

The definition below summarizes the definition of the module hierarchy. “The module hierarchy for a hierarchical Coloured Petri Net $CPN_H = (S, SM, PS, FS)$ is a directed graph $MH = (N_{MH}, A_{MH})$, where

- 1) $N_{MH} = S$ is the set of nodes.
- 2) $A_{MH} = \{s1, t, s2\} \in N_{MH} \times T_{sub} \times N_{MH} \mid t \in T_{sub}^{s1} \wedge s2 = SM(t)$ is the set of arcs.

The roots of MH are called prime modules, and the set of all prime modules is denoted SPM ” [39].

H. STATE SPACE ANALYSIS AND BEHAVIORAL PROPERTIES

To investigate the behavioral properties of the FMPR system a state space of the CP-Nets model is computed automatically. During model checking, temporal logics (i.e. LTL and CTL) are used for specification, checking, and verification of behavioral properties. These behavioral properties include the identification of the states in which the system terminates, the minimum and maximum number of tokens in a place, identification of states in which the system always reach (i.e. reach-ability analysis).

The CP-Nets toolkit named *cpntools* [40], [44] has a library for CTL model checking. In the case of the Hierarchical CP-Nets of FMPR the state spaces can be generalized by replacing places with place instances and transitions with transition instances. The *cpntools* tool-kit supports state spaces generation and analysis for hierarchical CP-Nets models of FMPR. A directed graph is constructed, having a node for each reachable marking and an arc for each occurring binding element.

I. MODERN PRODUCTION SYSTEMS - DISCRETE, CONTINUOUS, AND HYBRID PETRI-NETS

According to [45] modern production systems can be modeled as a continuous, discrete event, or hybrid systems. The Petri-nets formalism provides a framework and working paradigm for analysis, design, modeling, implementation, control, and operation of production systems. The Petri-nets in production systems improve the understanding of the

system and improve the communication between the stages of the life cycle.

Reference [46] have proposed the modeling of hybrid systems by using Petri-nets. Hybrid systems modeling is done by using continuous and discrete variables. Time may be either continuous or discrete. The continuous models can be extended to consider discrete aspects and the discrete models can be extended to include continuous aspects. Differential equations can be used for the continuous model of physical systems. The system state of the continuous model can be represented by continuous variables (real numbers) that are functions of time (i.e. time is continuous). CP-Nets represent systems by means of a discrete event model [46]. In a Discrete event model, time is discrete, and state variables are discrete.

In a Timed CP-Net the model is discrete and the continuous constructs are represented by specifying their time duration. The time duration is associated with places and transitions. The time duration is specified by real numbers (i.e. continuous values), and the CP-net model is discrete, thus this results in a hybrid model. The continuous variables are specified as real numbers (i.e. continuous place token count) and the discrete variables are specified as integers (i.e. discrete place token count) [46], [47].

The concept of continuous Petri-nets and hybrid Petri-nets were introduced in [48]. Hybrid Petri-nets were developed in [49]. According to [47] Petri-nets are ideal to model discrete event dynamic systems (i.e. computer systems, manufacturing systems, communication systems, etc). Continuous Petri-nets (i.e. markings are real numbers and the transition firings are continuous) can model a continuous system or approximate a discrete system. A hybrid Petri-net has a discrete as well as a continuous part.

According to [47] a hybrid Petri-net has both discrete and continuous parts and these parts are interconnected by arcs. The state-space analysis is performed by calculating the reachable states and verifying the properties of safety, liveness, progress, deadlock detection, and performance. In the case of a Petri-net with places having a large number of tokens, the number of reachable states explodes (i.e. state-space explosion). Autonomous continuous Petri-net is defined by [50]. A continuous Petri-net can be either autonomous (i.e. no time involved) or with firing speeds associated with transitions (i.e. time involved). To measure and evaluate the performance of the system the timed Petri-net model is ideal. A number of continuous Petri-net models have been defined having different calculations for measuring the time-duration (i.e. firing speed) of the transitions [48] [50]–[52].

Reference [53] has proposed a specialized CP-nets model for hybrid systems. In this model, the discrete timed Petri-net and a continuous model interact through event signals and conditions. This specialized hybrid net is a linear hybrid net condition/ event systems where the continuous part is a constant continuous Petri net. This specialized hybrid net has two methods for reach-ability analysis (1) *Predicate-transformation method*: an extension of state

reach-ability analysis for linear hybrid automata. (2) *Path-based method*: it verifies if a given set of states can be reached from another set by firing a sequence of discrete transitions and enumerates all possible firing sequences of discrete transitions.

Reference [54] proposed a model to serve as an analytical helping toolkit in short-term scheduling for the oil refinery process. Specialized Petri-net models are proposed for the special requirements and constraints of the oil refinery process. Reference [54] have proposed Controlled Colored Timed Petri-Net (CCTPN) to model the crude-oil operations in the oil refinery process. It is an extension of the hybrid Petri-Net. In CCTPN the continuous and discrete processes are integrated together. A token carries both discrete and continuous properties. A token acts as a discrete one in a discrete transition and a continuous one in a continuous transition. In a discrete place, the discrete properties of the token are shown, while in a continuous place the continuous properties are captured. Liveness property of CCTPN is defined, which identifies the conflicts in scheduling the system by using this model. The *process manufacturing system* is a hybrid system, the materials processed are continuous and are described by continuous variables, whereas the tasks start and end at discrete time points and are defined by discrete events.

Reference [55] has proposed a solution to the Short-Term Scheduling Problem (STSP) of crude-oil operations. As mathematical programming models do not meet the industry needs, [55] have proposed a formal model-based and control-theoretic method to solve this scheduling problem. The complex problem is broken down into tractable sub-portions. Each sub-portion has either discrete variables or continuous variables. A linear programming method is developed for the sub-portion with continuous variables. The heuristics method is used for the sub-portions with discrete variables. Therefore, a solution is developed for the STSP, and it is shown and validated via industrial case studies.

Reference [56] have scheduled a system in an oil refinery, having two pipelines to transfer different types of oils i.e. low-fusion-point crude oil, or high-fusion-point oil.

Reference [57] has proposed a solution to the short-term scheduling problem of the crude oil operations by using a combination of discrete-event and continuous processes. Hybrid Petri-net is used to specify the functional properties (i.e. behavior) of crude oil operations.

In our work, we have used *Hierarchical CP-Nets* for the multi-agent model of the FMPR. These CP-Nets have three abstraction levels and they are model-checked and the state space is generated and evaluated. In the future, we would use *Hierarchical Timed CP-Nets* for modeling the time-constraints of the FMPR system. In a Timed CP-Net, the continuous aspects are modeled in a discrete model by representing their time duration and the performance of the system is evaluated. The time duration is either associated with places or with transitions. These time-durations are continuous values (i.e. represented by real numbers), and the

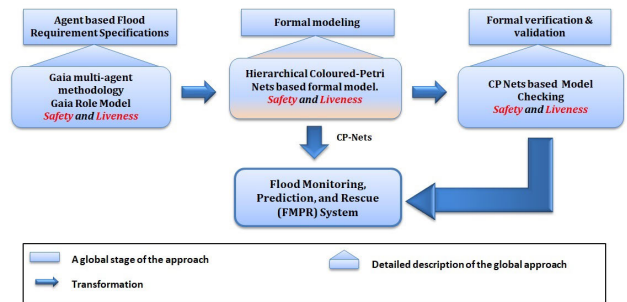


FIGURE 4. Proposed approach: multi-agent-based FMPR system.

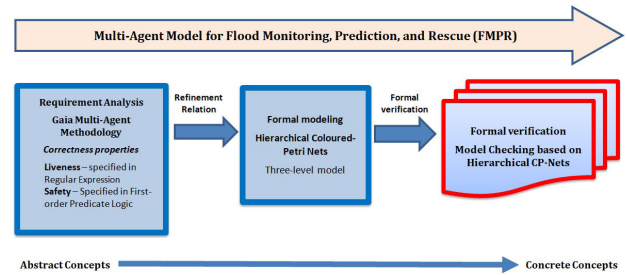


FIGURE 5. Requirement analysis, hierarchical CP-Nets-based modeling and model checking.

CP-nets model is discrete, thus, as a result, a hybrid model is obtained.

V. MATERIAL AND METHODS: MULTI-AGENT MODEL FOR FMPR SYSTEM

A. OBJECTIVES

One of the major objectives is to design and develop a correct, reliable, and complete model that ensures safety and liveness properties. A correct model is centered on four pillars: “(1) **Formal foundation**: Since the safety-critical FMPR system has several mandatory requirements, a sound formal foundation is a pre-requisite for the engineering approach. (2) **Executable language for specification**: Formally specifying a large-scale safety-critical system requires multiple models, with each model at its appropriate level of abstraction. Designing and realizing large-scale safety-critical systems requires suitable models at appropriate levels of abstraction (3) **Run-time execution platform**: Automatic adaptation and evolution of safety-critical systems requires a suitable run-time execution platform. (4) **Self-adaptation** as a technique to achieve resilience”.

A safety-critical FMPR system is a long-life system, therefore it should function in an open and dynamic environment. In a dynamic and open environment, the system can evolve, resources can change, the context of the system can change at any time, services may evolve, new services may be created. Figure 4 below shows the three global phases of the approach: (1) multi-agent-based requirement specification and design, (2) formal modeling, and (3) formal verification and validation.

Figure 5 below shows the formal methods and techniques that constitute the above global phases of the approach:

TABLE 2. Agent names, agent roles, and the description of these roles.

Agent Name	Agent Roles	Description
Sensor_Data Agent	Get Sensor Data Store Sensor Data	Collect real-time sensor data from sensors. Store the sensor data into the Data warehouse.
Flood_Perceiver Agent	Get Perceiver Status	Get the status of each of the flood perceivers.
Monitor Agent	Get Water Level Get Water Speed Calculate Threat level Inform Rescue Agent	Get real-time river water-level data and save it in the Data warehouse. Get real-time river water-speed data and save it in Data warehouse. Calculate the flood threat level by using the water-level, water-speed, and sensors data. Inform the rescue agent to provide rescue services before, during, and after a flood.
Rescue_Services Agent	Inform Rescue Services Activate Actuator Agent	Inform the different rescue services about the flood status and affected area before, during, and after floods. Activate the actuator agent to initialize the communication services, and rescue services.
Actuator Agent	Broadcast SMS Broadcast Radio Activate Rescue Drones	Activate the Actuator agent to send flood info SMS to affected areas population, before, during, and after floods. Activate the Actuator agent to send flood info via radio broadcast to affected areas population, before, during, and after floods. Activate the Actuator agent to send flood monitoring, prediction, and rescue drones to affected areas population, before, during, and after floods.
Prediction Agent	Process Data Give Recommendations	Huge datasets of sensor-data, sensor-logs, drones audio and video data (i.e. Big-data) stored in Data warehouse is processed using Machine Learning algorithms to make future predictions and future decisions. Huge datasets of sensor-data, sensor-logs, drones audio and video data (i.e. Big-data) stored in Data warehouse is processed using Machine Learning algorithms to make future recommendations

(1) Requirement analysis, preliminary and detailed design based on Gaia multi-agent methodology, (2) Hierarchical CP-Nets (i.e. multi-level nets) based formal modeling having multiple layers of abstraction, and (3) formal verification and validation based on exhaustive model checking of the CP-Nets.

B. ANALYSIS AND DESIGN: GAI-BASED ANALYSIS AND DESIGN OF FMPR SYSTEM

1) ORGANIZATIONAL RULES

Organizational rules specify the global rules and constraints of the system. These rules and constraints are identified by the analysis phase and are respected by the actual organization of agents. These constraints are satisfied and respected by agent roles and protocols. These organizational rules also define the environment of agents, and agents can perform autonomous actions in this environment. The environment mediates the interaction among flood monitoring, prediction, and rescue agents, and manages the access to flood data and information.

- 1) **Organizational Abstractions:** The overall un-structured and semi-structured data of the flood perceiver sensors, flood maps are placed in the database. The database of employees is an organized collection of data consisting of tables, queries, reports, and views.
- 2) **Multi-Agent Role Models:** Interface is an important component in the FMPR system, through this GUI the records from the database are accessed.

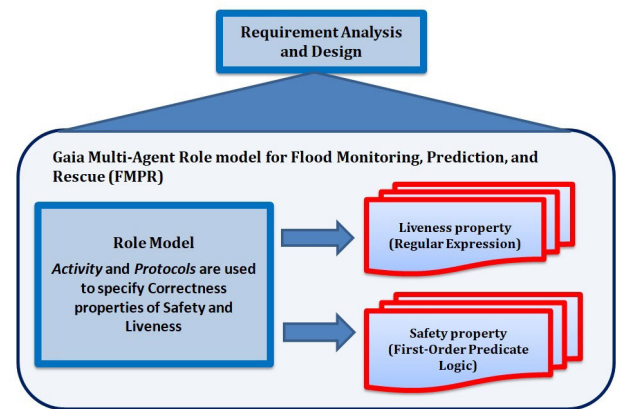
**FIGURE 6.** Requirement specification, design, and definition of correctness properties of liveness and safety.

Figure 6 below highlights the Gaia multi-agent methodology role model. The role model activities and protocols are used to specify the liveness and safety properties. The liveness property is specified in regular expression which is human readable, rigorous, precise, mathematically-founded and formal.

2) SYSTEM ROLE MODELS

The FMPR system is constituted by several agents. Each agent performs some specific tasks which are called the roles of the agent. An agent can have multiple roles. Below is

TABLE 3. Get sensor data: role of sensor_data agent.

Role Schema: Get Sensor Data
Description: Role of a Sensor_Data Agent
Protocols and Activities: <u>getFloodPerceiverID</u> , <u>getSensorID</u> , <u>connectToSensor</u> , <u>receiveSensorData</u> , <u>processSensorData</u> , <u>connectToMonitor</u> , <u>transferToMonitor</u> , <u>storeSensorData</u>
Permissions: reads: <i>sensor_id</i> , <i>sensor_data</i>
Responsibilities: Liveness: Get_SensorData = (<u>getFloodPerceiverID</u> . <u>getSensorID</u> . <u>connectToSensor</u> . Data_Processing . Monitor_Agent) Data_Processing = (<u>receiveSensorData</u> .(<u>processSensorData</u> <u>storeSensorData</u>)+) Monitor_Agent = (<u>connectToMonitor</u> . <u>transferToMonitor</u>)
Safety: <i>For all the sensors, sensors ID cannot be null and any two sensors cannot have the same sensor ID.</i> $(\forall x \in \text{sensor_id}(x): \neg(\text{sensor_id}(x) = \text{null})) \wedge (\exists x, y \in \text{sensor_id}: \neg(\text{sensor_id}(x) = \text{sensor_id}(y)))$

TABLE 4. Rescue operations role: role of the rescue_services agent.

Role Schema: Rescue Operations
Description: This role of the Rescue_Services Agent which involves managing the rescue operations before, during and after floods
Protocols and Activities: <u>getInfo</u> , <u>getVictimID</u> , <u>getVictimLocation</u> , <u>updateRecord</u> , <u>confirmUpdate</u>
Permissions: reads: <i>victim_id</i> changes: <i>victim_info</i> , <i>victim_status</i> (external)
Responsibilities: Liveness: Update = (<u>getVictimID</u> , <u>getVictimLocation</u> , <u>getNoOfVictim</u> , <u>updateRecord</u> , <u>confirmUpdate</u>)
Safety: <i>For each victim, the victim ID cannot be null and any two victims cannot have the same victim ID and</i> <i>For each victim, the victim location cannot be null and for all victims, in case of a rescue operation, the number of victims cannot be null and any two victims cannot have the same information (i.e. name, father name, NIC #, etc.)</i> $(\exists x \in \text{victim_id}(x): \neg(\text{victim_id}(x) = \text{null})) \wedge (\exists x, y \in \text{victim_id}: \neg(\text{victim_id}(x) = \text{victim_id}(y)))$ $\wedge$ $(\exists x \in \text{victim_location}(x): \neg(\text{victim_location}(x) = \text{null})) \wedge (\forall x \in \text{victim_count}(x): \neg(\text{victim_count}(x) = \text{null}))$ $\wedge$ $(\exists x, y \in \text{victim_id}: \neg(\text{victim_info}(x) = \text{victim_info}(y)))$

the table highlighting the agent names, their roles, and the description of these roles.

A. Get Sensor Data Role (shown in table 3) is the role of the Sensor_Data Agent. Its objective is to collect data from flood sensors and then store this data into the Data warehouse.

As shown in table 3, activities (i.e. underlined) are connectToMonitor and transferToMonitor and protocols are getFloodPerceiverID, getSensorID, connectToSensor, receiveSensorData, processSensorData, storeSensorData. These protocols define the mapping between two agent roles. The safety property ensures that the sensor_id is not null, and is unique (i.e. no two sensors are having the same id).

B. Rescue Operations Role is the role of the Rescue_Services Agent. Its objective is to manage the rescue operations.

As shown in table 4, there is only one activity updateRecord and protocols are getVictimID, getVictimLocation, getNoOfVictim, and confirmUpdate. The safety property ensures that the victim_id is not null, and is unique. It updates the information of the victim.

C. HIERARCHICAL CP-NETS MODEL AND VERIFICATION: MULTI-AGENT MODEL OF FMPR

A hierarchical CP-Nets model for FMPR multi-agent system is specified and modeled. It consists of places, transitions, and labeled arcs between places and transitions. It performs model checking i.e. exhaustive state-space evaluation of the system. CP-Nets has an integrated specification language called CP-Nets ML, which is a functional meta-language. It is formal, precise, well-defined, and unambiguous. CP-Nets ML has a concept of color sets as Data-types. New color sets can be easily defined, declared and initialized. The liveness and safety properties defined in earlier analysis, design, and detailed design phases are now transformed in the form of hierarchical CP-Nets, for the formal verification and validation.

The Hierarchical CP-Nets model is organized as a set of modules. This makes a modular design with three levels of abstractions. Therefore, it has a better model architecture having better debugging, easy upgrades, a high degree of human-machine interaction (HMI). These modules are



FIGURE 7. CP-Nets colour sets data types of the communication model of FMPR system.

- 1) Communication module.
- 2) Flood Perceivers i.e. flood sensors module.
- 3) Control unit module.
- 4) User Interface module.
- 5) Rescue Services module.

A complete hierarchical CP-Nets model of the FMPR system communication services is constructed specifying all the sensors, actuators, and control units of the system. The proposed communication model has several composite color sets that represent complex data structures. This composite color set uses (i) Cartesian product of two or more color sets (ii) Union (iii) List and (iv) Record. The primitive, as well as composite data-types (i.e. color sets) of the system, are shown in Figure 7.

Each flood sensor has a unique ID. Color set S_ID stands for Sensor ID and its datatype is an integer, Color set S_DATA stands for Sensor data and its datatype is String. Color set $S_ID \times S_DATA$ is a composite data-type representing a Cartesian product of S_ID and S_DATA (i.e. a tuple consisting of Sensor ID and its data). Color set $DATAPACKS$ is a list of $S_ID \times S_DATA$ (i.e. it can store a large number of data packets with each packet consisting of sensor ID and its data). Color set $ACKPACKS$ stands for acknowledgment packets is a list of sensor id. When a data packet reaches the destination from the source than an acknowledgment is received. In this way, the loss of packets in the network as well as duplicate packets is avoided.

Figure 8 shows the first level of the hierarchical CP-Nets model of Communication between flood perceivers (i.e. input sensors) and the control unit of the system. This model consists of places, transitions, and the data is represented by composite color sets (i.e. Cartesian product and list). The model is distributed and if some part of the system fails (i.e. if some data packets are lost) the system ensures that duplicate packets are generated and the information is transferred without any loss. The control unit sends an acknowledgment packet as soon as it receives a data packet. As a result, there is no loss of information between flood monitoring sensors and the control unit.

Figure 9 shows the input portion of the model. It specifies two flood monitoring perceivers. These flood monitoring perceivers have flood monitoring sensors that provide real-time sensor data as input to the system. Each data packet has two parts, the first part is the sensor id and the second part is the sensor data. This composite data packet is represented as a Cartesian product of Sensor ID and Sensor Data (i.e. $S_ID \times S_DATA$). These data packets are stored in a composite data structure called list, a queue of these data packets are formed, and each packet is sent to the network portion, one after another (i.e. First-In-First-Out).

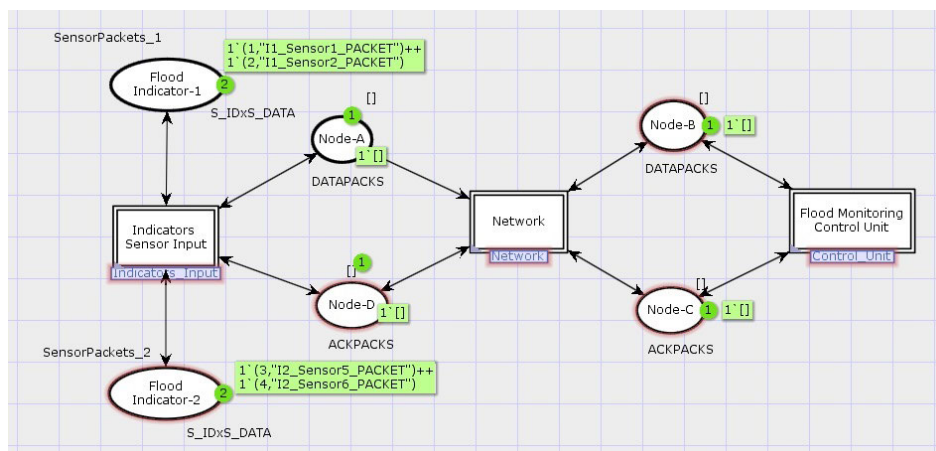


FIGURE 8. Global CP-Net for the FMPR system communication model.

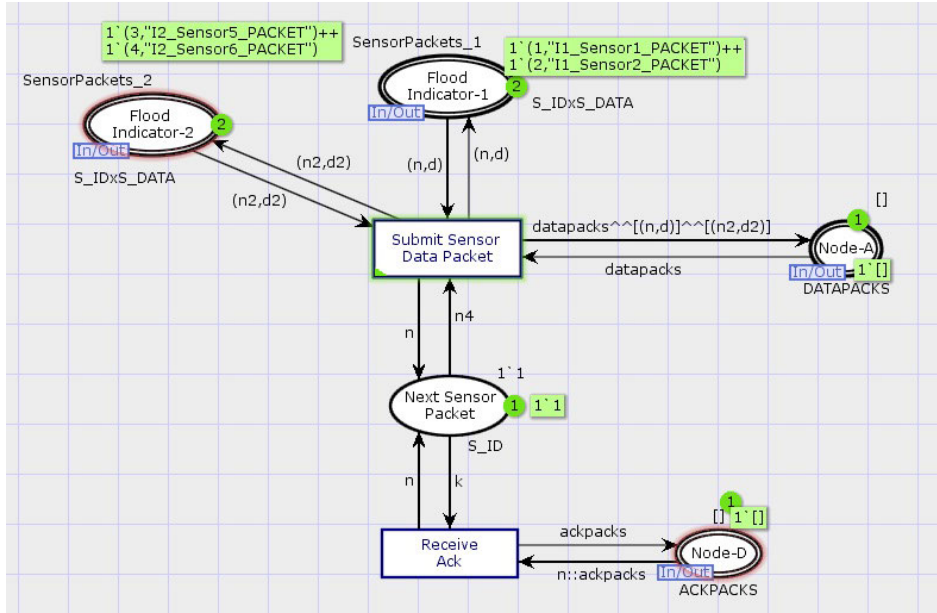


FIGURE 9. CP-Net input model for the FMPR system communication model.

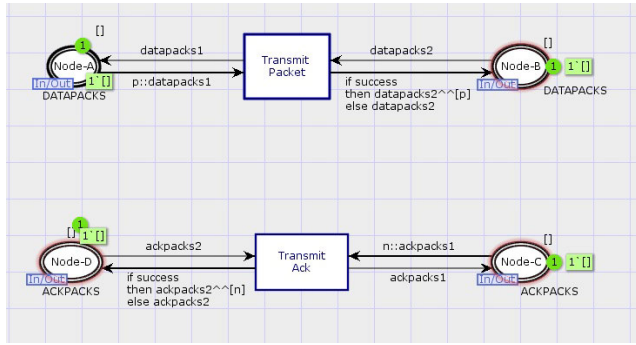


FIGURE 10. CP-Net network model for the FMPR system communication model.

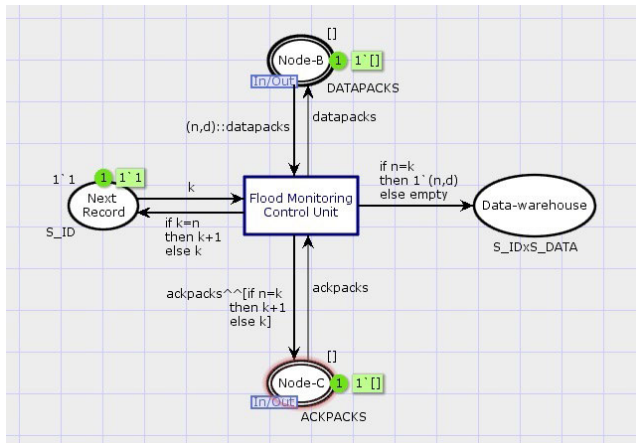


FIGURE 11. CP-Net control unit for the FMPR system communication model.

The network portion of the model is shown in Figure 10 ensures that the system does not get congested and dead-locked. This network model has two parts, the first part

transfers the data packets from the flood monitoring perceivers to the control unit, and ensures that there is no loss of data packets. The second part sends an acknowledgment for each data packet received.

The control unit of the system is shown in Figure 11. It receives the data packets and transfers them for storage towards the data-warehouse. It also discards the duplicate data packets so that the data redundancy can be reduced. It also manages and coordinates with the network part, so that no data packet is lost and data packets are received in their correct sequence.

VI. RESULTS, DISCUSSION, AND CONCLUSION

This work results in a correct, reliable, formally designed and modeled agent-based safety-critical FMPR system. Our previous expertise on ensuring correctness properties of liveness and safety for safety-critical systems like multi-agent robotic system [58]–[60], and FMPR systems [61]–[63] have greatly helped us during the formal analysis, design, modeling, and verification of this system.

The FMPR system is based on multi-agents. The system is fault-tolerant and it works even if several agents fail. It has a Gaia based formal requirement specification and design. The multi-agent analysis and design of the system are completed by using Gaia's role model-based formal specification of liveness and safety properties. The liveness properties are specified formally by using regular expression and safety properties are specified by using first-order predicate calculus. FMPR is formally modelled and model-checked by using *Hierarchical CP-Nets*. The *formal verification of correctness properties* is carried out by using a Hierarchical CP-Nets based model checking. The CP-Nets hierarchical model precisely specifies and defines the correctness properties of liveness and safety. This model is a state transition machine with

states and transitions, where the data-flow is represented by tokens. A token can represent primitive data-types, composite data-types, and composite data-structures. This model performs rigorous model-checking. The Hierarchical CP-Nets are constructed with two abstraction levels (i.e. layers). As a result of this two-level refinement (i.e. from abstract to concrete concepts), the model of the flood monitoring system is modular (i.e. broken down into well-defined sub-parts). This modular system has a high degree of flexibility, fault-tolerance, and human-machine interaction. It also helps in exhaustive model-checking. The system provides *emergency and rescue services* after a flood. It proposes several communication mediums (i.e. radio, internet, social media, electronic media, mobile network, etc.). The system is centered on *uninterrupted continuous communication* between the FMPR system and stakeholders (i.e. affected citizens, social media, rescue services, law and order authorities, electronic media, and government disaster management authorities). Propagation of information to these stakeholders ensures early action after flood alerts. CP-Nets model of communications is constructed and model-checked.

In this work, we have studied discrete, continuous, and hybrid Petri-nets and we have found *Hierarchical CP-Nets* ideal to develop a multi-agent model of the FMPR. These CP-Nets are then model-checked using toolkit cpntools. In the future, we have planned to use *Hierarchical Timed CP-Nets* for the modeling and verification of FMPR system. We are looking forward to using timed CP-Net, in order to model time and to specify and ensure *performance* properties. As performance is directly related to time durations of the firing of tokens in CP-Nets. More specifically, in our future work, we are looking forward to formally specify, model, and verify our system by using Hybrid timed CP-Nets. Our nets would have both discrete and continuous parts interconnected by arcs. The CP-net model would be discrete but it would have markings of real numbers and continuous transition firings. It would have timing speeds associated with transitions (i.e. time constraints). Therefore, resulting in a hybrid model.

The findings of this work clearly show that a methodology based on formal methods and techniques can be used for the analysis, design, modeling, and verification of multi-agent safety-critical systems like the FMPR system. Formal methodology like Gaia provides specialized models for the requirement specification, analysis, organizational abstractions, design, and detailed design of multi-agent-based systems. Hierarchical CP-Nets provide precise and rigorous modeling and model checking.

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